Causal Attribution Under Confounding

Revised project direction: the ML contribution carries the novelty, and the visualization displays it

Course: Advanced Data Visualization Techniques
Domain: Business / Marketing Analytics
Status: Revised direction, replaces the earlier "Touchpoint Constellation" ML plan

1. Why the Earlier Plan Was Weak

The earlier pitch used XGBoost, Shapley, Markov, SHAP, k-means, and bootstrap. All of these are off-the-shelf and common in attribution work. Stacking them adds length, not novelty, and the visualization alone cannot be the novelty. This version puts the contribution in the ML.

2. The Real Problem

Every standard attribution method (last-click, linear, Markov, Shapley) measures **correlation**. They split credit among the channels present in converting journeys. They cannot tell whether a touchpoint *caused* the conversion or merely appeared for someone who was going to buy anyway.

This matters in practice:

- Retargeting and branded search are shown mostly to people who are already close to buying.
- Every correlational method rewards those channels for that, so budget flows to channels that harvest demand rather than create it.
- What the marketer actually needs is **incrementality**: how many conversions happened *because* of the channel. Correlational methods cannot estimate it.

3. The ML Core: Causal, Overlap-Aware Attribution

Task: estimate the *incremental* effect of each channel touch on conversion, given that exposure was not random.

Method (one primary estimator, not a menu of models):

- Treat each touchpoint as a time-varying treatment, with the journey history up to that point as the confounders.

- Learn a propensity model for "why did this user receive this exposure now?", using gradient boosting or a GRU over the journey history.
- Use a sequential doubly robust estimator (or a stabilized inverse-probability-weighted marginal structural model) to estimate each channel's incremental conversion effect.

The proposed contribution: honest uncertainty about what the data cannot identify

- Where the propensity is near 0 or 1 (a channel fires almost only for high-intent users), there is no comparable unexposed group. **Overlap diagnostics** flag these channels.
- A **sensitivity analysis** lets the user state how strong hidden intent might be, and the credit interval widens accordingly.
- **Output per channel:** an incremental credit interval plus an identifiability flag, instead of a falsely precise single number.

4. Validation Plan

1. **Simulator with known ground truth.** Hidden intent drives both exposure (for example retargeting) and conversion. The true incremental effect of each channel is set by the project author.
2. **Benchmark.** Compare last-click, Markov, Shapley, and the causal estimator against the known truth. The expected result is that standard methods over-credit retargeting and the causal estimator moves closer to the truth.
3. **Interval coverage check.** Verify that the intervals contain the true effect at roughly the claimed rate. If they do not, the method is wrong, and that gets reported.
4. **Real data run.** Apply the method to the Criteo attribution dataset, which has covariates, cost, and click fields usable for propensity modeling. There is no ground truth here, so use negative-control channels as placebo tests and report where the data is non-identifiable.

5. What Is and Is Not Novel

The individual estimators (marginal structural models, doubly robust methods, sensitivity analysis) already exist, and causal attribution is an active research area. It is not certain that this exact combination is unpublished.

The defensible claim is:

> Overlap-aware causal credit with sensitivity-bounded intervals for multi-touch journeys, benchmarked against standard attribution on a simulator with known truth.

Before the word "novel" appears anywhere in the report, spend about an hour on a literature check (for example, existing causal multi-touch attribution papers) and position the work against them.

6. What to Cut

Item	Decision
SHAP replay panel	Cut, it does not answer the causal question
k-means journey clusters	Cut
LSTM or attention attribution	Cut
Story mode and decorative effects	Cut
Markov and Shapley	Keep only as baselines to compare against
Budget simulator	Keep only if it uses the causal estimates with intervals, otherwise it reintroduces the correlation problem

7. Where the Visualization Fits

The visualization displays what the ML produces. It is not claimed as the novelty.

Visual element	Encodes
Star size	Causal incremental credit
Halo width	Sensitivity-bounded interval
Hidden-intent slider	Widens or narrows the halos as the user changes the assumed confounding strength
Correlational vs. causal comparison view	Per channel, shows credit under standard methods against causal credit, so over-credited channels are obvious
Identifiability flag	Marks channels where overlap is too poor to estimate an effect

8. Data

Dataset	Role
Custom simulator	Primary benchmark with known ground truth
Criteo Attribution Modeling for Bidding dataset	Real-data application, campaign-level, requires accepting Criteo's terms and citing the paper

9. Risks

- **Causality cannot be proven from observational data.** The honest claim is "less biased, with stated limits, validated on simulation." Avoid the phrase "true effect" on real data.
- **The simulator is the main cost.** A lazy simulator makes the benchmark meaningless, so confounding must be realistic. Budget about one day.
- **Three days is tight.**
- **Novelty claim risk.** Related causal attribution work exists, so the claim must be framed carefully after the literature check.

10. Priority Order for 2 to 3 Days

1. Simulator with known ground truth
2. Baseline attribution models (last-click, Markov, Shapley)
3. Causal estimator with overlap diagnostics
4. Interval coverage check and sensitivity analysis
5. Real-data run on Criteo with placebo tests
6. Visualization of the results

11. Deliverables

- Simulator and benchmark notebook
- Causal attribution pipeline with documented assumptions
- Interactive visualization of causal credit, intervals, and identifiability
- Short report covering the problem, method, validation results, limitations, and literature positioning